

Geophysical Inversion

Homework 1

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Problem 1.4

Let the data $\mathbf{d}$ be the running average of the model parameters, $\mathbf{m}$, computed by averaging groups of three neighboring points; that is, $d_i = (m_{i-1} + m_i + m_{i+1})/3$

A

What is the matrix $\mathbf{G}$ in the equation $\mathbf{d} = \mathbf{G}\mathbf{m}$ in this case?

Solution

The matrix $\mathbf{G}$ is the matrix with zeros almost everywhere except around the diagonal where the number $\frac{1}{3}$ appears three times (See section B for details).

B

What problems arise at the top and bottom rows of the matrix and how can you deal with them?

Solution

The definition of the matrix runs into problems at the boundaries where either index, $i - 1$ or $i + 1$ are out of bounds and the corresponding parameter doesn't exist. There are two straightforward solutions.

Possibility 1

We take the average of the available parameters, giving us:

$$\begin{aligned}d_1 &= (m_1 + m_2)/2 \\d_N &= (m_{N-1} + m_N)/2\end{aligned}$$

Where N is the total number of parameters, which will be equal to the total number of datapoints for this example. This yields the matrix:

$$\mathbf{G} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & \dots & \dots & \dots & \dots & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & & & & & \vdots \\ 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & & & & & \vdots \\ \vdots & & & & \ddots & & & & \vdots \\ \vdots & & & & & & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \\ \vdots & & & & & & 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & \dots & \dots & \dots & \dots & 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

Possibility 2

We simply ignore the boundary. This is for example often done in image processing when applying a convolutional filter. This means that $\mathbf{d}$ will have 2 fewer elements than $\mathbf{m}$ which might not be desirable. This approach would yield the matrix

$$\mathbf{G} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & \dots & \dots & \dots & \dots & 0 \\ 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & & & & & \vdots \\ \vdots & & & & \ddots & & & & \vdots \\ \vdots & & & & & & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \\ 0 & \dots & \dots & \dots & \dots & 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}$$

C

How sparse is $\mathbf{G}$? What percent of it is zero?

Solution

The precise percentage depends on the solution to the boundary problem but it will be around $\frac{N-3}{N} \times 100\%$